

INTELLIPAAT CAPSTONE ASSIGNMENT

NAME – EDWARD GHOSH

Q1. Mr. A who is an HNI has the following long position in equity market:

1. Purchased Ashok Leyland: 5,000 units @ ₹180 each

2. Purchased L&T: 300 units @ ₹2,600 each

Mr. A is seeking your assistance to hedge the exposure in the equity market by using Futures and Options.

Q1. Perform hedging strategy using Futures and Options separately. (10 Marks)

Solution:

A. Hedging Using Futures

1. Ashok Leyland Futures

Futures Lot Size: 7,000 shares (NSE standard)

To hedge 5,000 shares, he can sell 1 lot (7,000) of Ashok Leyland Futures.

Selling futures offsets loss if stock price falls.

Action:

Sell 1 lot Ashok Leyland Futures (7,000 units) at current futures price (assume ₹182)

Though he holds 5,000 shares, he'll slightly over-hedge with 7,000 futures.

2. L&T Futures

Futures Lot Size: 300 shares

He owns exactly 300 shares.

Action:

Sell 1 lot L&T Futures (300 units) at current futures price (assume ₹2,620)

Result of Futures Hedge:

If market falls, Mr. A loses in spot shares but gains on short futures.

If market rises, Mr. A gains in shares but loses on short futures.

Net position remains stable — this is a perfect hedge, but profits are capped.

B. Hedging Using Options

1. Ashok Leyland Put Options

Buy 1 Put Option (strike price ₹180) for 7,000 shares (1 lot)

Premium (cost): Assume ₹4 per share → Total = ₹28,000

This gives Mr. A the right to sell at ₹180, limiting downside.

Action:

Buy 1 lot Ashok Leyland Put Option (₹180 strike)

2. L&T Put Options

Buy 1 Put Option (strike price ₹2,600) for 300 shares (1 lot)

Premium: Assume ₹70 per share → Total = ₹21,000

Action:

Buy 1 lot L&T Put Option (₹2,600 strike)

Result of Options Hedge:

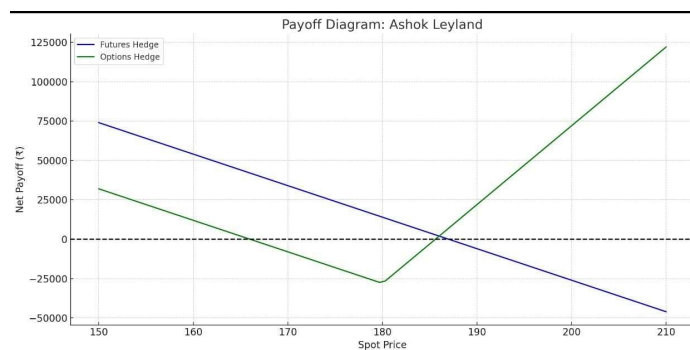
If market falls, value of Put increases → limits losses

If market rises, shares gain in value, and he loses only the premium

Q2. Make a pay-off diagram in both Futures and Option cases.

Solution:

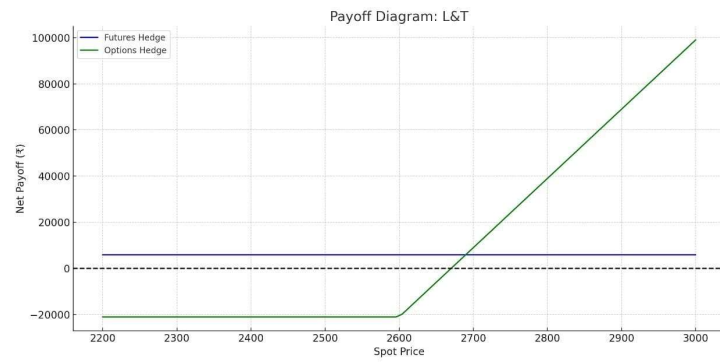
1. Ashok Leyland



Blue Line (Futures Hedge): Your profit/loss is fixed. Gains from stock are cancelled by losses on futures, and vice versa.

Green Line (Options Hedge): You are protected if price falls below ₹180, but you can still enjoy profits if the price goes up — you only lose the premium (₹4/share).

2.L&T



Futures Hedge (Blue Line): Your gains and losses are fully neutralized. There is no profit or loss overall.

Options Hedge (Green Line): You are protected from a fall below ₹2,600. If price rises, you still enjoy profits after subtracting the premium (₹70/share).

Q3. Calculate breakeven point and profit and loss using option strategy. (10 Marks)

Solution:

Step-by-Step Option Strategy

1. Ashok Leyland

Bought: 5,000 shares at ₹180

Bought 1 Put Option (Lot size = 7,000) at ₹180 strike

Paid Premium = ₹4 per share → Total = ₹28,000

2. L&T

Bought: 300 shares at ₹2,600

Bought 1 Put Option (Lot size = 300) at ₹2,600 strike

Paid Premium = ₹70 per share → Total = ₹21,000

Breakeven Point (BE)

Formula:

Breakeven = Purchase Price + Premium Paid

Ashok Leyland

Purchase Price = ₹180

Premium = ₹4

Breakeven = ₹184

L&T

Purchase Price = ₹2,600

Premium = ₹70

Breakeven = ₹2,670

Profit and Loss Scenarios

Let's check 3 cases:

1. Price falls below strike
2. Price stays at strike
3. Price goes above breakeven

Ashok Leyland Example:

Market Price	Profit/Loss Explanation	Net Result
Rs. 60	Stock Loss = $(160-180) * 5000 =$ Rs. 100000	Put Gain = $(180-160) * 7000 =$ Rs.140000. Total = Rs. 140000- 100000-28000 = Rs. 12000 Profit
Rs. 180	No Gain/Loss in stock or put	Loss = Only Premium = Rs.28000
Rs. 190	Stock Gain = $(190-180) * 5000$ = Rs. 50000 Put = 0	Net = Rs. 50000-Rs. 28000 = Rs. 22000 Profit

L&T Example:

Market Price	Profit/Loss Explanation	Net Result
Rs. 2500	Stock Loss = $(2500-2600) * 300$ = -Rs. 30000	Put Gain = $(2600-2500) * 300$ = Rs. 30000 Total: Rs. 30000-30000-210000 = Rs. 21000 Loss (Only Premium paid)
Rs. 2600	No Gain/ Loss in the stock or put	Loss = Only Premium = Rs. 21000
Rs. 2700	Stock Gain = Rs. 30000 Put = 0	Net = Rs. 30000-21000 = Rs. 9000 profits

Summary Table :

Stock	Breakeven Price	Max Loss	Upside Potential
Ashok Leyland	Rs. 184	Rs. 28000 (premium)	Unlimited
L&T	Rs. 2670	Rs. 21000 (premium)	Unlimited

Conclusion:

Mr. A uses Put Options to protect from losses below ₹180 and ₹2,600.

His maximum loss is the premium paid.

He can still earn profit if stock price goes up — only breakeven is shifted higher.

Q4. Q4. Perform calculations using Excel and write a detailed process in Word document. (10 Marks)

Stock	Units	Purchase Price (₹)	Total Investment (₹)
Ashok Leyland	5,000	180	9,00,000
L&T	300	2,600	7,80,000
Total			16,80,000

Stock	Lot Size	Futures Price (Approx.)
Ashok Leyland	7,000	₹182
L&T	300	₹2,620

2. Hedging Using Futures:

To hedge against downside risk:

- **Sell Futures** equivalent to or close to current holding.

Ashok Leyland:

- Mr. A holds 5,000 shares, but 1 futures lot = 7,000
- Hedge with **1 futures lot** (slightly over-hedged)

L&T:

- Mr. A holds 300 shares
- 1 lot = 300 → Hedge with **1 futures lot**

3. Hedging Using Put Options:

Alternatively, buy Put Options to limit downside.

Ashok Leyland:

- Buy 1 Lot (7,000) of ₹180 Strike Price Put Option

L&T:

- Buy 1 Lot (300) of ₹2,600 Strike Price Put Option

This protects the downside below the strike price while allowing profit if the stock price rises.

Strategy	Cost Involved	Protection	Upside Participation
Futures (Short)	No premium, margin required	Full below futures price	No upside gain
Options (Put)	Premium paid	Below strike	Unlimited upside

Conclusion:

Mr. A can use either **Futures (for full hedge)** or **Options (for limited risk hedge)** depending on his risk appetite. The assignment illustrates how to apply derivatives for portfolio protection.